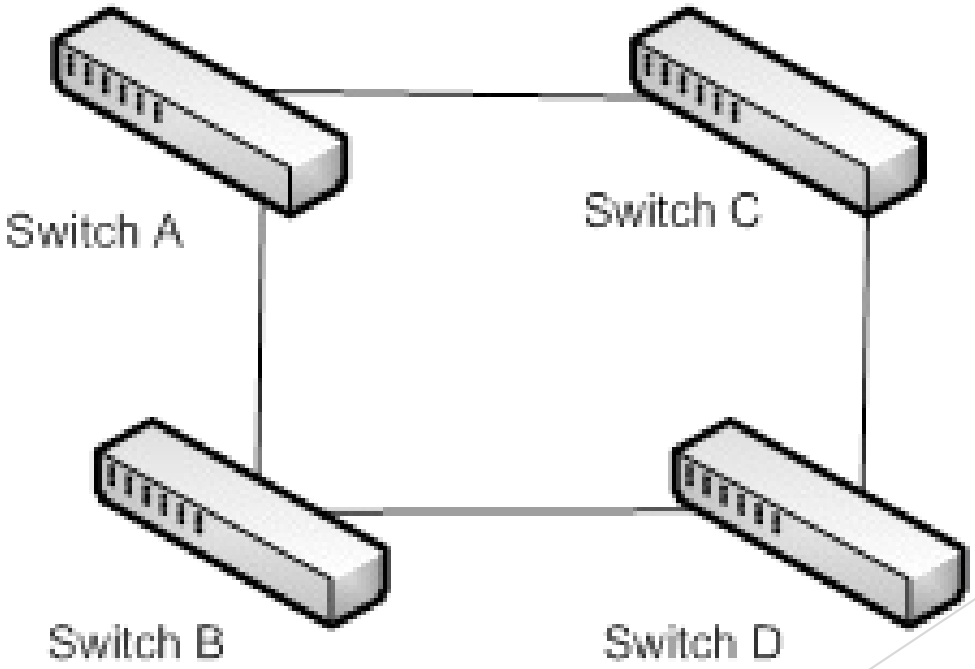
Spanning Tree Protocol
Port Security
BPDUguard

NETC-121 Week 9

STP

- Spanning Tree Protocol
  - Removes layer 2 loops. (It is a protocol just like TCP/IP is a protocol)



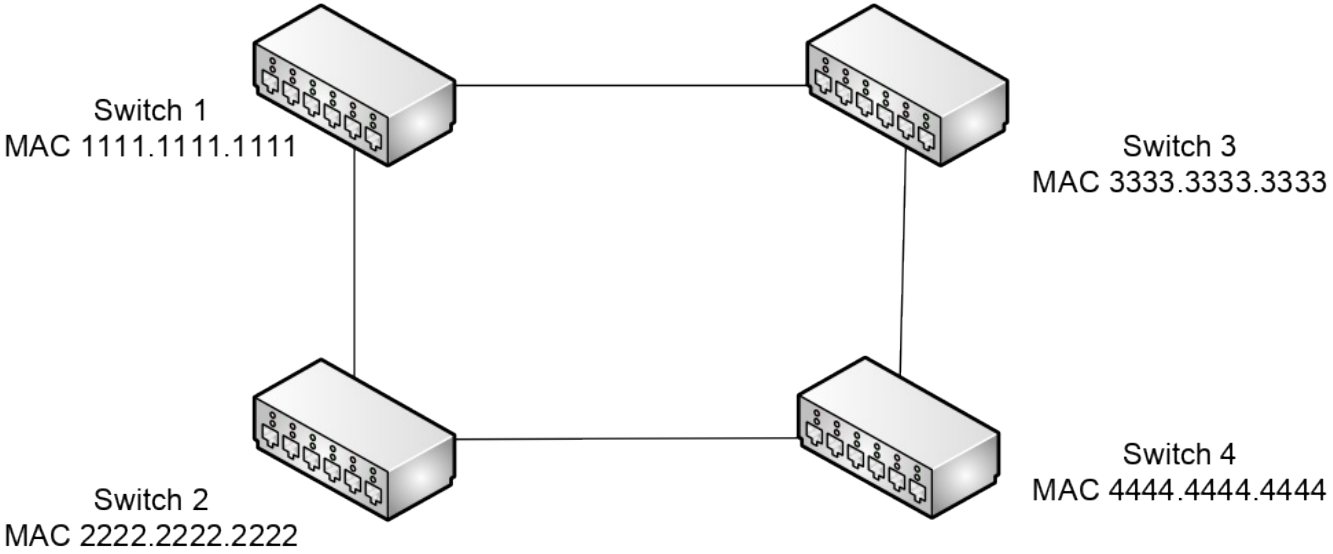
BPDU

- Bridge Protocol Data Unit
  - How switches share Layer 2 information
  - Sent out as Multicast every 2 seconds
  - Contains switch ID
    - Switch ID is a switches priority value and MAC address
    - Default priority value is 32768
    - Can change a switches priority value
    - Example: switches MAC address is 1111.2222.3333
      - Switch ID would be 32768-1111-2222-3333

Root Switch

- Root Bridge – The switch with the lowest switch ID is the root switch.
  - The root switch makes all of the layer 2 looping decisions.
  - On election, each switch calculates and comes up with the same switch as being the root.
    - Every time the topology changes, the election process starts over. There is one root switch per LAN.

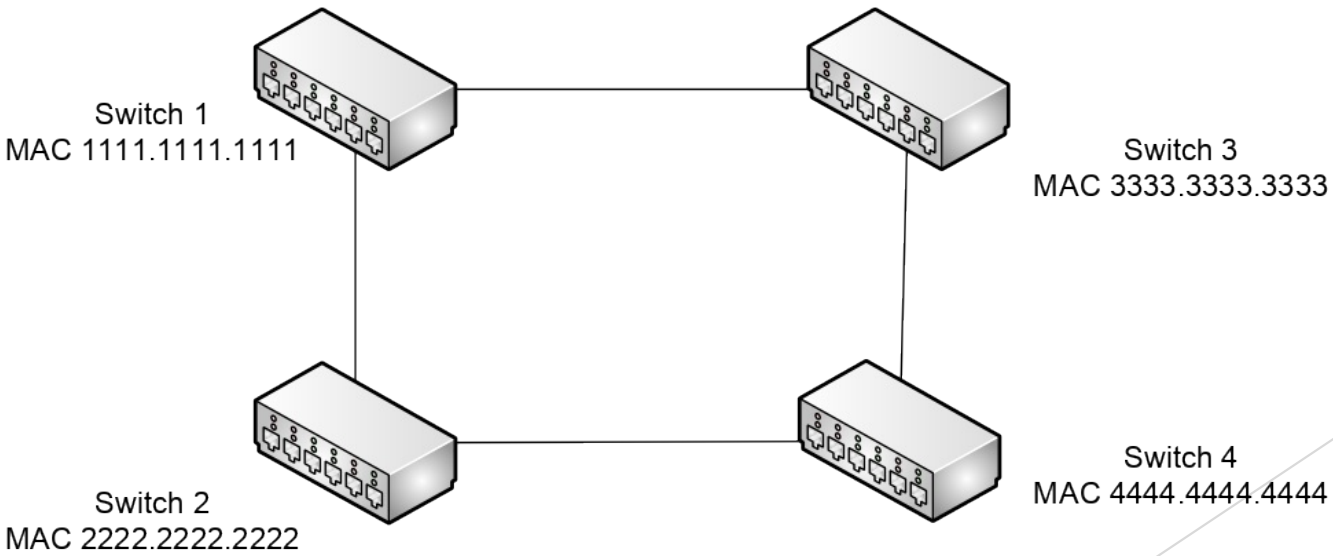
Root Switch


Switch 1
MAC 1111.1111.1111
Switch 3
MAC 3333.3333.3333
Switch 2
MAC 2222.2222.2222
Switch 4
MAC 4444.4444.4444

Root Switch

- In the diagram below Switch 1 would become the root switch because it has the lowest switch ID
  - Switch ID: 327681111.1111.1111

- If we wanted switch 3 to be the root we would change its priority value (20000)
  - Switch ID: 200003333.3333.3333

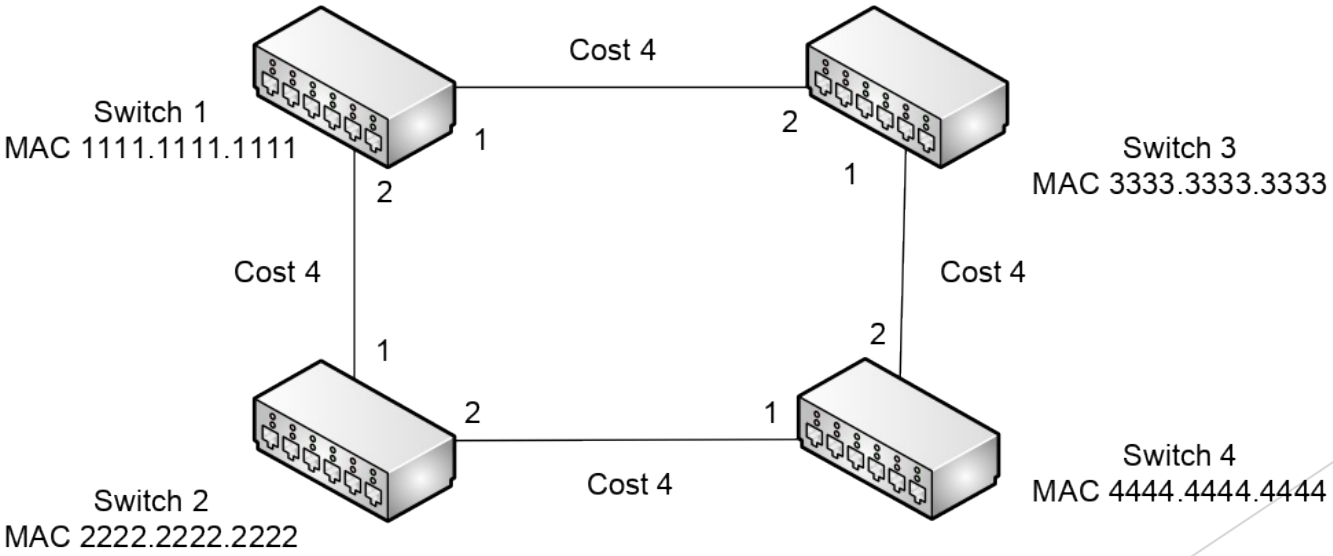


Root Port

- Each switch that is not the root port selects a port that will be used to reach the root switch.
- Root port is selected based on the following:
  - Based on lowest path cost to the root switch (Cost is inverse bandwidth, Higher cost = low bandwidth. Low cost = higher bandwidth).
  - Lowest neighboring switch ID
  - Lowest port priority value
  - Lowest physical port number

Root Port

- Switch 1 - None, it is the root
- Switch 2 - Port 1 (cost is 4) to root switch
- Switch 3 – Port 2 (cost is 4) to root switch
- Switch 4 – Port 1 (cost is 8 both ways but port 1 is connected to the neighbor with the lowest switch ID (32768.2222.2222.2222 vs 32768.3333.3333.3333)



STP

- When a new device is plugged in
  - A switch does not know if it is another switch
  - It could be a PC, printer, router, new switch…
  - Switch goes through port states to identify what type of device is connected
  - Determines if there is a new root or a loop

STP Port States

- Blocking – 20 seconds, Port only listening for and processing BPDU's, all other traffic dropped
  - Port may remain in this state if it prevents a loop
- Listening - 15 seconds, Double checking topology, only processing BPDU's
- Learning - 15 seconds, Processing BPDU's and updating CAM table but not forwarding frames
- Forwarding – Normal switch port operation
- Disabled – Not participating in STP, manually shutdown or nothing plugged in

STP Timers

- Forward Delay Timer
  - How long STP waits between listening and learning states
  - Can be changed to decrease STP convergence time
  - Default is 15 seconds

- Hold Down Timer
  - How long the host remembers the root switch
  - STP root switch sends BPDU's every 2 seconds
  - Timer is set for 6 seconds so after 3 missed BPDU intervals it is assumed the root switch is down
    - A new root switch election would occur
    - This prevents a topology with no root switch
  - Every time a device receives a BPDU from the root switch the timer resets to zero

STP Cont.

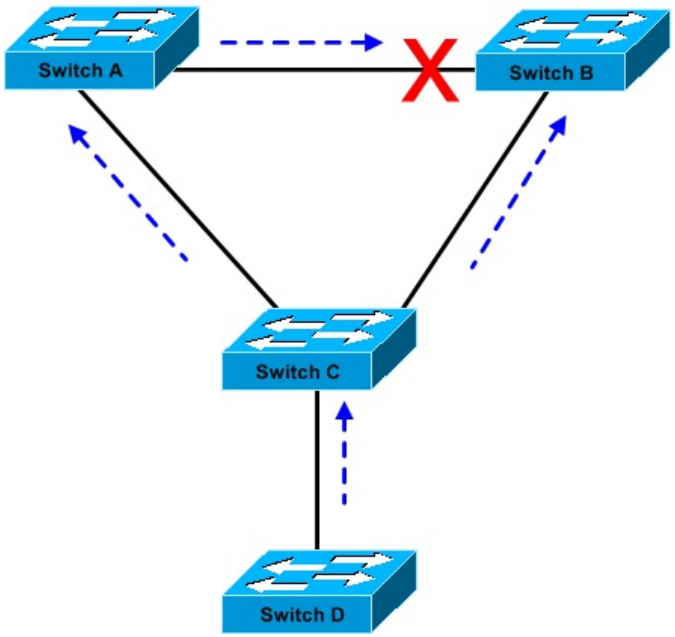
- Ports take 50 seconds to go through STP process.
  - That's why switch ports are orange for 50 seconds when you plug a new device in.
  - During that time the port will not forward traffic
- Convergence – when all ports and devices are at the normal operating state
- To view STP information on a switch – switch# show spanning-tree-protocol

Port Fast

- *interface fa 0/1*
  - *spanning-tree portfast*

- Allows the port to immediately transition to Forwarding Mode, Skipping Listening and Learning
- Typically for *Edge* devices (Not Switches / Routers)
- Can allow for LOOPS to happen

BPDUGuard

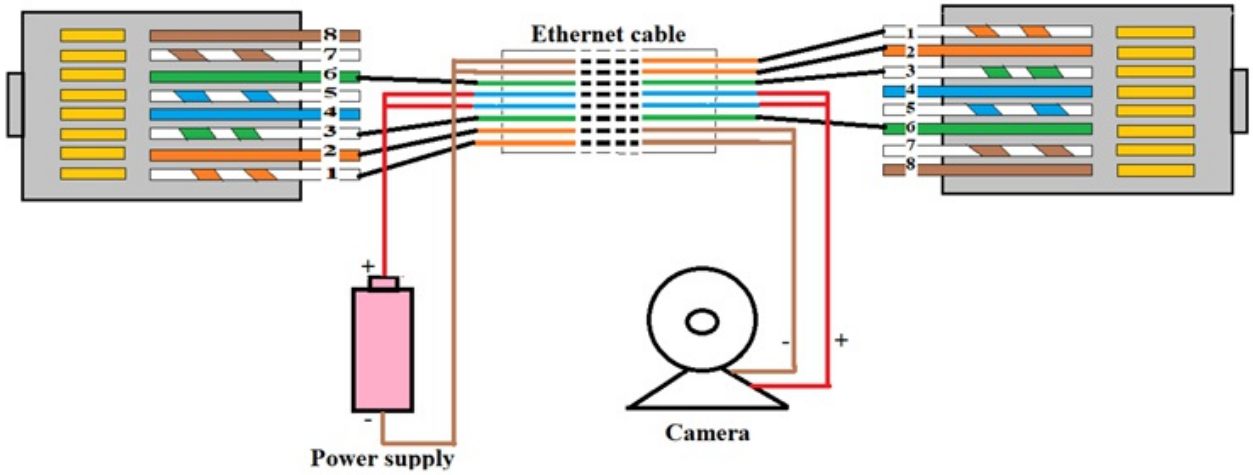


- Spanning Tree PortFast and BPDU Guard has been Enabled
- Computer is connected to a port on Switch C
- Computer is no longer connected to port on Switch C
- Switch D is NOW connected to port on Switch C
- Switch D sends BPDU to Switch C
- Switch C says “I shouldn’t receive a BPDU on this port”
  - Shutdown Port to Switch D – *Error Disabled*

Lab Commands

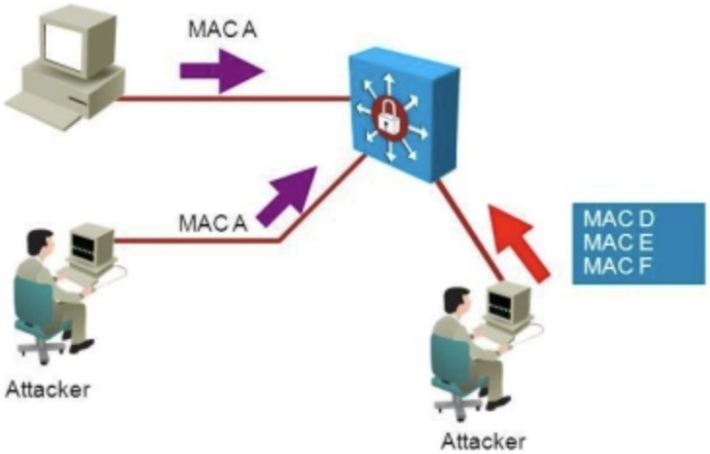
- *show spanning-tree*
  - Shows just about everything you need for STP
- *spanning-tree vlan 1 priority 0*
  - Changes priority for vlan 1 to 0, which is the highest priority. Your priority number must be a multiple of 4096. (0, 4096, 8192, etc.)
- *show spanning-tree interface fa 0/1*
  - Shows spanning tree info for a specific port

Power over Ethernet (PoE)



- In addition to Data Communication, Power is provided across the same cable to power the device on the other side. This could be a
  - IP Phone
  - IP Camera
  - Wireless Access Point
  - Door Buzzer
  - Emergency Call Box
  - Etc…
- Power can be provided via a PoE Capable Switch or via a PoE Injector

Port Security




```
Cat3550(config)# interface gigabitethernet 0/5
Cat3550(config-if)# switchport mode access
Cat3550(config-if)# switchport port-security
Cat3550(config-if)# switchport port-security maximum 5
Cat3550(config-if)# switchport port-security violation protect
Cat3550(config-if)# switchport port-security mac-address 1234.1234.1234
Cat3550(config-if)# switchport port-security mac-address sticky
Cat3550(config-if)# end
```

```
Switch# show port-security interface fa0/20
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 3 mins
Aging Type                 : Inactivity
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : 0009.7C10.8E8C:50
Security Violation Count   : 1
```